

Started on Wednesday, 28 August 2024, 3:11 PM

State Finished

Completed on Wednesday, 28 August 2024, 4:01 PM

Time taken 50 mins 43 secs

Grade 80.00 out of 100.00

Question **1**

Correct

Mark 20.00 out of 20.00

Write a Python Program to evaluate the series:

$1/1! + 1/2! + 1/3! + \dots + 1/n!$ using recursion.

For example:

Input	Result
4	1.7083333333333335

Answer: (penalty regime: 0 %)

```

1 def factorial(n):
2     if n==0:
3         return 1
4     return(n*factorial(n-1))
5 limit=int(input())
6 sum=0.0
7 for i in range(1,limit+1):
8     sum=sum+(1/factorial(i))
9 print(sum)

```

	Input	Expected	Got	
✓	4	1.7083333333333335	1.7083333333333335	✓
✓	7	1.7182539682539684	1.7182539682539684	✓
✓	10	1.7182818011463847	1.7182818011463847	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **2**

Correct

Mark 20.00 out of 20.00

1. A. Write a Python Program to convert a decimal number to a binary number using tail recursion.

For example:

Input	Result
12	1100

Answer: (penalty regime: 0 %)

```

1 | def decimal_binary(n):
2 |     if n==0:
3 |         return 0
4 |     else:
5 |         return (n%2)+10*decimal_binary(int(n/2))
6 | n=int(input())
7 | print(decimal_binary(n))
8 |

```

	Input	Expected	Got	
✓	20	10100	10100	✓
✓	12	1100	1100	✓
✓	36	100100	100100	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question 3

Incorrect

Mark 0.00 out of 20.00

Write a python program to evaluate the expression for n=10 using recursion.

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n \text{ for } |x| < 1$$

For example:

Input	Result
0.8	0.6032774143999999

Answer: (penalty regime: 0 %)

```

1 def fun(x,n):
2     if (n==0):
3         return 1
4     else:
5         return(((x**n))+fun(x,n-1))
6 x=float(input())
7 n=10
8 print(fun(x,n))

```

	Input	Expected	Got	
✗	0.8	0.6032774143999999	4.5705032704000015	✗
✗	0.6	0.6272674816000001	2.4909300736000004	✗
✗	0.5	0.6669921875	1.9990234375	✗
✗	0.4	0.7143156735999998	1.6665967616000004	✗

Your code must pass all tests to earn any marks. Try again.

Show differences

Incorrect

Marks for this submission: 0.00/20.00.

Question 4

Correct

Mark 20.00 out of 20.00

Write a program to count the consonants in a string using recursion

For example:

Input	Result
tree	2

Answer: (penalty regime: 0 %)

```

1 def isConsonants(ch):
2     ch=ch.upper()
3     return (not (ch == 'A' or ch == 'E' or ch == 'I' or ch == 'O' or ch == 'U'))
4
5 def totalConsonants(string,n):
6     if n==1:
7         return isConsonants(string[0])
8     return totalConsonants(string,n-1)+isConsonants(string[n-1])
9 string=input()
10 print(totalConsonants(string,len(string)))

```

	Input	Expected	Got	
✓	tree	2	2	✓
✓	four	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.

Question **5**

Correct

Mark 20.00 out of 20.00

Define the abstract base class named Polygon and also define the abstract method. This base class inherited by the various subclasses. Implement the abstract method in each subclass. Create the object of the subclasses and invoke the **sides()** method.

For example:

Result
Triangle has 3 sides
I have 4 sides
Pentagon has 5 sides
Hexagon has 6 sides

Answer: (penalty regime: 0 %)

Reset answer

```

1 | from abc import ABC
2 |
3 | class Polygon(ABC):
4 |
5 |     # abstract method
6 |     def sides(self):
7 |         pass
8 |
9 | class Triangle(Polygon):
10 |
11 |
12 |     def sides(self):
13 |         print("Triangle has 3 sides")
14 |
15 | class Pentagon(Polygon):
16 |
17 |
18 |     def sides(self):
19 |         print("Pentagon has 5 sides")
20 | class Hexagon(Polygon):
21 |
22 |     def sides(self):

```

	Expected	Got	
✓	Triangle has 3 sides I have 4 sides Pentagon has 5 sides Hexagon has 6 sides	Triangle has 3 sides I have 4 sides Pentagon has 5 sides Hexagon has 6 sides	✓

Passed all tests! ✓

Correct

Marks for this submission: 20.00/20.00.